The various commercially available Li-ion batteries are lithium-manganese oxide (LMO), lithium-iron phosphate (LFP), lithium-nickel manganese cobalt oxide (NCM), lithium-nickel-cobalt-aluminium oxide (NCA), and lithium-titanium oxide (LTO). Each of these has its own pros and cons. Savio Monteiro, Senior Vice President - Energy, Oil & Gas and Utilities, Praxis Global Alliance, informs that the NCM batteries NCM622 and NCM811 are improving the battery technologies in terms of energy density and life cycle.

Shedding light on the roadblocks hampering the progress of battery technologies in India, Shibulal says, "Thermal management is the biggest roadblock. The weather conditions and temperature constraints are important aspects that manufacturers keep in mind. The temperature in India goes as high as 48 to 50°C, but the batteries available today are suitable for only up to 40°C."

High temperature causes the liquid electrolyte between the cathode and anode to expand, putting stress on outer casings and making them prone to bulging, cracking, or even puncture. So, they need a porous barrier to limit chemical activity. Moreover, extreme heat causes combustion readily, so as the temperature increases further, there is a possibility of explosion or at least degradation of the battery capacity. EV batteries, therefore, must be equipped with cooling systems, which makes them more expensive.

Fortunately, some newer designs such as solid-state and sodium-ion batteries are safer. No separators are required in these, leading to space saving and shrinking battery sizes. Solid-state batteries also have a longer lifespan. However, there is a huge amount of work that needs to be done to make sure these batteries are viable for the commercial market.

Extensive research is going on to develop graphene and metal-air batteries. Shibulal affirms that hybrid ultracapacitor batteries are also being developed, which would use supercapacitors for fast charging while maintaining the energy density of the batteries.

Naresh Neelkantan, Senior Architect, Sasken Technologies, says about the future of battery technologies in India, "The raw material supply security is an important parameter in the context of a sustainable battery manufacturing ecosystem. India does not have reserves of lithium, added to which there is heavy dependency on imports for cobalt and nickel as well. These are critical raw materials for some front-runner lithium-ion battery systems." He informs, India is gradually breaking into the sectors of mining. Going ahead, the growth in the resources for battery manufacturing will play a major role in the development of an ecosystem in India.

Battery demand

Ashwin Shankar, CEO, BatteryPool, believes that unlike the technologies of the past, the demand for batteries will majorly come from the Indo-Pacific region which has a combination of fast-growing and energy-hungry economies. A lot of countries in this region are looking for energy security in the long term. In India, we have policies and subsidies driving the adoption of battery technologies, both from the supply side and the demand side.

The automotive industries require batteries of high performance at elevated temperatures as well as high capacity with more range and life-cycle, making current automobile OEMs take a more proactive approach towards the research and development of solid-state batteries. Due to continuous electrochemical processes happening inside a battery due to charging and discharging, after a few hundred charge-discharge cycles, the battery's internal resistance increases significantly, and the battery can no longer be used for applications such as electric vehicles.

Ashwin points out an interesting aspect of the batteries used in electric vehicles. When these degrade and are left with only 70% state of health (SoH), becoming unsuitable for use in EVs, the batteries can be used for backup power in grids. This creates room for new value chains around batteries.

The biggest consumers of batteries currently are the consumer electronics and mobility sectors. Renewable energy systems are gaining traction to meet energy demands and lowering emissions. Complimenting that, we need batteries to store that energy and